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Assignment-Regression Algorithm
Insurance Charges Prediction

1. **Identify your problem statement.**
   Stage 01 = Since the data deals with categorical and numerical data its falls under Machine Learning domain.
   Stage 02 = The requirement is clear which is predict the charges of the policy and all data present in the dataset, so it belongs to Supervised learning method.
   Stage 03 = Since the target variable is numerical/continuous value we choose the Regression.

2. **Tell basic info about the dataset (Total number of rows, columns).**
   Total number of rows = 1338 rows
   Total number of columns = 6
   It has [‘age’, ‘sex’, ‘bmi’, ‘children’, ‘smoker’, ‘charges’] columns.

3. **Mention the pre-processing method if you’re doing any (like converting string to number – nominal data)**

   The columns SEX and SMOKER has the categorical data, needs to do the pre-processing step before passing to the model.

4. **Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.**

   Please refer notebook for coding.

DECISION TREE

CRITERION	SPLITTER	R2_SCORE
absolute_error	best	0.661034699
absolute_error	random	0.699609348
friedman_mse	best	0.685759409
friedman_mse	random	0.728699614
absolute_error	best	0.670750428
absolute_error	random	0.689973867
poisson	best	0.711014117
poisson	random	0.697962765

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RANDOM FOREST	
N_ESTIMATORS	R2_SCORE
10	0.833030413
25	0.850765958
50	0.849832932
75	0.852461078
100	0.853830791
200	0.852477748

5. **All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)**

#	ALGORITHM	R2_SCORE
1	Multiple Linear Regression	0.789479035
2	SVM	-0.111661287
3	DecisionTree (friedman_mse + random)	0.728699614
4	RandomForest (n_estimators = 100)	0.853830791

6. **Mention your final model, justify why u have chosen the same**.
Overall the RandomForest's model perform well and as of this data set SVM gives poor performance.

So we save the RandomForest's model (with n_estimator=100) and will deploy for user's access.

Ex: 42 years old non-smoking male, having 2 kids, cost will be 8,439.72

Deployment

```
import pickle
filename = "finalized_model_Insurance.sav"
```

```
pickle.dump(regressor, open(filename, "wb"))
```

```
loaded_model = pickle.load(open(filename, "rb"))
```

```
result = loaded_model.predict([[42, 22.4, 2, 0, 0]])
print("Predicted cost for 42 yrs, 22.4 BMI, number of kids = 2, Male and Non Smoker is", result)
```

```
Predicted cost for 42 yrs, 22.4 BMI, number of kids = 2, Male and Non Smoker is [8439.72816425]
```